Individual Coursework Submission Form

Specialist Masters Programme

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INTRODUCTION

This report examines the distribution of EO50 certificates throughout New York City boroughs over time, taking into account socio-economic factors to offer insights into ICAP. The focus of this analysis is primarily on the boroughs of NYC and economic variables, utilizing three main datasets:

1. eo50.csv contains data about businesses that have received the EO50 certificate.

2. Borough_boundaries.geoJSON obtained from NYC open data, which provides the geographical boundaries of NYC's boroughs, minus water areas.

3. Nyc_census_tracts.csv sourced from Kaggle, containing socio-economic and demographic data.

By integrating these three datasets, this report seeks to offer business and policy insights into how EO50 practices differ across various locations in NYC based on two plots, with particular attention to borough-level data and economic indicators. Two visualizations have been created to illustrate these plots:

1. A spatial distribution map showing EO50 certificates across NYC Boroughs, with employment rates indicated by colour and poverty rates shown by point size.

2. A temporal diffusion plot displaying the evolution of EO50 certificates by borough from 2021-2024, with median income shown by point size.

VISUALIZATIONS

I. Geo-Spatial Diffusion of EO50 Certificate across New York City's Boroughs

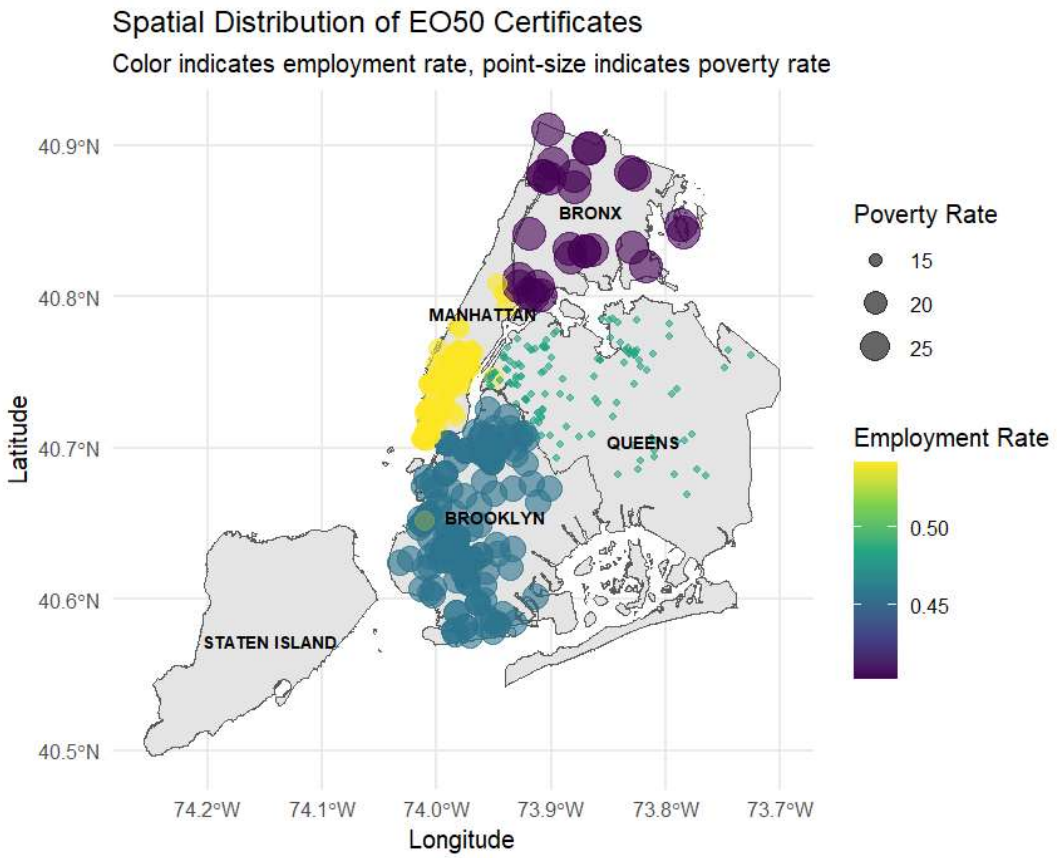


Fig (i): Spatial Distribution of EO50 Certificates Across NYC Boroughs. Employment rates shown by colour gradient and poverty rates by circle size.

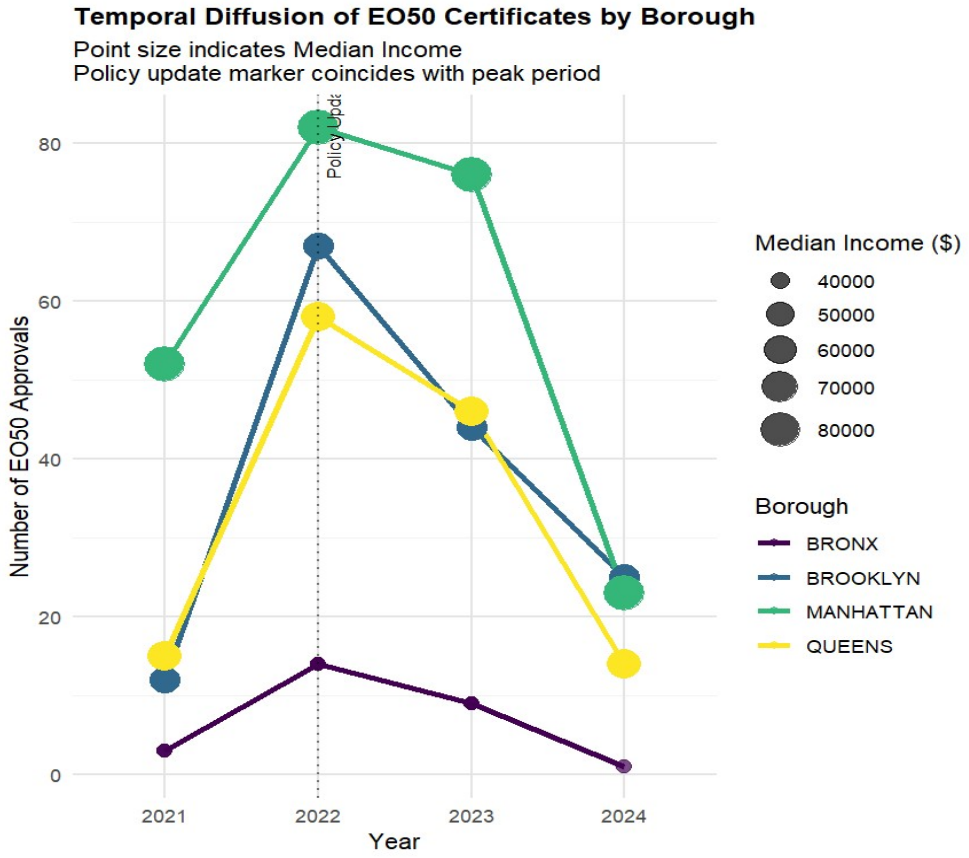
Business/Policy Insights

The above Fig(i) illustrates the distribution of the EO50 certificate across the boroughs of New York City, with colour indicating employment rates and point size reflecting poverty rates. The choice of employment and poverty rates as economic indicators is based on their direct connection to ICAP's objectives of economic growth and job creation, as well as their focus on equal opportunity rights for individuals. It was found that the Boroughs with higher employment rates tend to show greater EO50 compliance. An inverse relationship was found between poverty rates and certificate density.
The main insights derived from the graph are as follows:

1. Manhattan shows a high density of EO50 certificate approvals, accompanied with employment rate above 50% and a lower poverty rate (indicated by smaller point sizes).
2. Queens displays a moderate density of EO50 certificate approvals, a moderate employment rate, and a lower poverty rate (also shown by smaller point sizes).
3. Brooklyn has a moderate density of EO50 certificate approvals but exhibits the employment rate between 45%-50% and a higher poverty rate (denoted by larger point sizes).
4. The Bronx has a lower density of EO50 certificate approvals, a lowest employment rate below 45%, and a higher poverty rate (represented by larger point sizes).

5. Staten Island presents a unique case in certificate landscape. The borough's minimal certification rates can be attributed to several factors, including its geographical isolation from the rest of New York City.

II. Temporal Diffusion of EO50 Certificates by Boroughs over time (2022-2024)



Fig(ii): Annual Trends in EO50 Approvals by Borough (2021-2024), with point size indicating median income levels.

Business/Policy Insights

The above Fig(ii) demonstrates the progression of EO50 approvals over time for each borough, where median income is represented by the size of the points. The choice of income as the economic indicator was made because it sheds light on the economic health of each borough and its potential link to ICAP participation. It is observed that higher median income areas demonstrate more sustained compliance over time.
The key trends observed from the graphs are:

1. 2022 Approval Spike – A significant increase in EO50 approvals across all boroughs was observed in 2022. This spike may signify that businesses are placing a greater emphasis on compliance with equal employment standards.
2. Manhattan's Dominance – Manhattan consistently has the highest number of approvals, likely due to its dense concentration of businesses. Additionally, the geo-spatial plot indicates that Manhattan has a higher employment rate and lower poverty levels.
3. Declining trend post-2022 shown by all boroughs indicating potential policy implementation challenges.

4. Staten Island Lacking Signs – There is minimal EO50 certification in Staten Island compared to other boroughs. This could be attributed to several reasons, including a lower business density compared to other NYC boroughs and less participation in the ICAP program.

COMPANION TEXT

One important step in New York City's efforts to advance economic and business development while guaranteeing workplace equality is the ICAP and EO50 policies that go along with it. Important trends that should be carefully considered for policy enhancement and implementation are shown by the examination of the distribution of EO50 certificates.

The spatial plot (Fig(i)) illustrates a notable disparity in EO50 compliance across NYC's Boroughs. The large concentration of businesses in Manhattan, especially in areas where employee rates are higher than 50%, indicates that established companies have had greater success enforcing equitable employment policies. Stronger institutional capability and increased resource availability in these places could be the cause of this. However, Bronx has the greatest employee and poverty rates and the lowest certificate density, suggesting a possible gap in the policy's reach. This discrepancy calls into question the effectiveness of the ICAP and EO50 programs as well as their reach in communities with low economic status. A balance in implementation is provided by Brooklyn and Queens, which exhibit moderate to high compliance levels with differing rates of poverty.

A clear trend in certificate issuance among boroughs can be seen in the temporal graph. The policy coincided with a spike in certificate issuance in 2022, indicating that changes in policies can have a significant impact on a company. However, the subsequent drop in all boroughs after 2022 suggests that firms are having trouble keeping up the trend and small businesses faced increased challenges in maintaining regulatory compliance due to limited resources and complex requirements. According to Manhattan's projections (highest median income levels and highest level of certifications), financial resources are essential to preserving compliance.

Going forward, there are opportunities as well as obstacles in the interaction between economic conditions and the EO50 certificate. Greater compliance is seen in regions with high employment rates and median incomes, which suggests stable economies and improved application of equal opportunity laws. To fix these disparities, NYC Department of Small Business Services (SBS) needs to help the businesses in the boroughs that are struggling economically and spread more awareness amongst the businesses.

The analysis provides several recommendations that NYC Department of SBS can implement:

1. Targeted Outreach: NYC Department should focus on areas with lower compliance rates, mainly Bronx and outer borough areas by introducing specific initiatives through seminars and workshops.

2. Resource Allocation: Department can firstly, implement a dedicated fund for the lower-income area businesses, secondly, implement a tiered support system based on business size and income levels providing more intensive assistance to smaller businesses.

3. Economic Integration: Provide businesses who maintain consistent compliance over multiple years with tax incentives.

4. Policy Sustainability: Instead of annual evaluations, implement quarterly compliance check-ins to address the problems early.

5. Focus on Outer-Boroughs- As seen in the visualizations, the problem of low participation of businesses in Staten Island can be addressed by implementing a specific outreach program in collaboration with local business associations could increase awareness and participation in EO50 certification. Secondly, creating transportation-linked incentives, such as discounted ferry passes for compliant businesses, could help overcome the borough's geographical barriers. Lastly, proposing additional incentives for businesses could attract industries that typically have high compliance rates thereby boosting the borough's overall certification numbers.

Finally, ICAP and EO50 programmes have tried to introduce the policies in some boroughs of NYC, but the disparities in implementation suggests the need for a nuanced approach that considers maintaining the equal opportunity practices, resource availability, and policy awareness amongst the businesses. NYC Department of SBS can build a more resilient and equitable business ecosystem that serves all communities regardless of their socio-economic status and geographic location.